

Foundation (Jalapeno)

- Q1.** What is the x -coordinate of the intersection point of $y = 2x + 1$ and $y = x - 2$?
(A) -3
(B) -1
(C) 1
(D) 3
(E) 5
- Q2.** If $y = 3x - 2$ and $y = x + 4$, what is the y -coordinate of the intersection point?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10
- Q3.** Graph the system $x + y = 4$ and $x - y = 2$. What is the intersection point?
(A) (1, 3)
(B) (2, 2)
(C) (3, 1)
(D) (4, 0)
(E) (0, 4)
- Q4.** Solve the system $2x + 3y = 7$ and $x - 2y = -3$ by graphing. What is the x -coordinate of the intersection point?
(A) -1
(B) 1
(C) 2
(D) 3
(E) 4
- Q5.** A stock's price is modeled by $y = 2x + 5$ and its competitor's price is modeled by $y = x + 10$. At what x -value do the two stocks have the same price?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q6.** Solve the system $y = x - 1$ and $y = -2x + 3$ by graphing. What is the y -coordinate of the intersection point?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q7.** Graph the system $y = x + 2$ and $y = -x - 4$. What is the intersection point?
(A) $(-2, 0)$
(B) $(-1, -1)$
(C) $(0, 2)$
(D) $(1, 3)$
(E) $(2, 4)$
- Q8.** Solve the system $x + 2y = 6$ and $3x - 2y = 4$ by graphing. What is the x -coordinate of the intersection point?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Open Ended Questions (FRQ)

Q9. Tom has 1200 to invest in two accounts. Account A earns 2% interest and account B earns 5% interest. If Tom invests x dollars in account A and

- Q10.** A bakery sells large and small cakes. The large cakes require 2 hours to bake and 5 hours to decorate, while the small cakes require 1 hour to bake and 2 hours to decorate. If the bakery has 10 hours to bake and 15 hours to decorate per day, write the system of linear inequalities to model this situation and graph the system to find the possible combinations of large and small cakes the bakery can produce.

Chef's Tips

- Tip 1: Remind students to label the x and y axes when graphing systems of linear equations.
- Tip 2: Encourage students to check their intersection points by plugging the values back into the original equations.
- Tip 3: Suggest that students use different colors to graph each equation in the system.